

R Code

```
1  # Practical 4 : Hotelling's Two-Sample T2 Test
2  #
3  # Dataset : iris
4  #####
5
6  # install.packages("Hotelling")
7
8  #####
9  # Load Package
10 #####
11
12 library(Hotelling)
13
14 #####
15 # Load Dataset
16 #####
17
18 data(iris)
19
20 #####
21 # Display First Six Observations
22 #####
23
24 head(iris)
25
26 #####
27 # Select Two Species
28 #####
29
30 setosa <- subset(iris, Species == "setosa")[, 1:4]
31
32 versicolor <- subset(iris, Species == "versicolor")[, 1:4]
33
34 #####
35 # Sample Sizes
36 #####
37
38 cat("\nSample Size of Setosa      :", nrow(setosa), "\n")
39 cat("\nSample Size of Versicolor :", nrow(versicolor), "\n")
40
41 #####
42 # Mean Vectors
43 #####
44
45 cat("\nMean Vector of Setosa\n")
46 print(colMeans(setosa))
47
48 cat("\nMean Vector of Versicolor\n")
49 print(colMeans(versicolor))
50
```

```
51 #####
52 # Covariance Matrices
53 #####
54
55 cat("\nCovariance Matrix of Setosa\n")
56 print(cov(setosa))
57
58 cat("\nCovariance Matrix of Versicolor\n")
59 print(cov(versicolor))
60
61 #####
62 # State the Hypotheses
63 #####
64
65 cat("\n-----\n")
66
67 cat("\nNull Hypothesis (H0):\n")
68 cat("The population mean vectors of Setosa and Versicolor are equal.\n")
69
70 cat("\nAlternative Hypothesis (H1):\n")
71 cat("The population mean vectors of Setosa and Versicolor are different.\n")
72
73 cat("\n-----\n")
74
75 #####
76 # Perform Hotelling's Two-Sample T2 Test
77 #####
78
79 result <- hotelling.test(setosa, versicolor)
80
81 #####
82 # Display Test Result
83 #####
84
85 cat("\nHotelling's Two-Sample T2 Test Result\n")
86 print(result)
87
88 #####
89 # Extract and Display P-value
90 #####
91
92 pvalue <- result$pval
93
94 cat("\nP-value =", format(pvalue, scientific = TRUE), "\n")
95
96 #####
97 # Decision Rule
98 #####
99
100 alpha <- 0.05
101
102 if (pvalue < alpha) {
103
104     cat("\nDecision: Reject the Null Hypothesis (H0)\n")
105 }
```

```
105
106   cat("\nConclusion:\n")
107   cat("There is sufficient statistical evidence at the 5% significance level\n")
108   cat("to conclude that the population mean vectors of Setosa and\n")
109   cat("Versicolor are significantly different.\n")
110
111 } else {
112
113   cat("\nDecision: Fail to Reject the Null Hypothesis (H0)\n")
114
115   cat("\nConclusion:\n")
116   cat("There is insufficient statistical evidence at the 5% significance level\n")
117   cat("to conclude that the population mean vectors of Setosa and\n")
118   cat("Versicolor are significantly different.\n")
119 }
120
121 #####
122 # Practice Question
123 #####
124
125 # Using the mtcars dataset, perform Hotelling's Two-Sample  $T^2$  Test
126 # to compare automatic and manual transmission cars based on:
127 #
128 # 1. Miles per Gallon (mpg)
129 # 2. Horsepower (hp)
130 # 3. Weight (wt)
131 #
132 # State the null and alternative hypotheses, obtain the test
133 # statistic and p-value, and give the conclusion at the 5%
134 # significance level.
135 #####
```